Валидация методов отбора данных, основанных на геометрических свойствах внутренних представлений

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Школа Анализа Данных

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Цель

Исследовать методы отбора данных для обучения языковых моделей, основанные на геометрических свойствах внутренних представлений, и проверить гипотезу о том, что такие метрики, как внутренняя размерность (Intrinsic Dimensionality) и Effective Rank, позволяют находить более информативные примеры.

Intrinsic Dimensionality (MLE)

Оценка внутренней размерности по расстояниям до ближайших соседей:

$$\hat{d} = \left(\frac{1}{k-1} \sum_{i=1}^{k-1} \log \frac{T_k}{T_i} \right)^{-1}$$

где:

T_i — расстояние до i-го ближайшего соседа,

T_k — расстояние до k-го соседа.

Intrinsic Dimensionality (TwoNN)

Отношение расстояний до ближайших соседей:

$$\mu_i = \frac{T_{2,i}}{T_{1,i}}$$

Оценка размерности:

$$d = \left(\frac{1}{n} \sum_{i=1}^{n} \log \mu_i \right)^{-1}$$

где:

$T_{1,i}$, $T_{2,i}$ — расстояния до ближайших соседей.

Correlation Dimension (CorrInt)

Корреляционный интеграл:

$$C(\varepsilon) = \frac{2}{n(n-1)} \sum_{i<j} \mathbb{I}(\|x_i - x_j\| < \varepsilon)$$

Оценка размерности:

$$d = \lim_{\varepsilon \to 0} \frac{\log C(\varepsilon)}{\log \varepsilon}$$

Effective Rank

Нормировка сингулярных значений:

$$p_i = \frac{\sigma_i}{\sum_j \sigma_j}$$

Эффективный ранг:

$$r_{\text{eff}} = \exp\left(-\sum_i p_i \log p_i\right)$$

где:

σ_i — сингулярные значения.

Schatten Norm

Определение нормы:

$$\|X\|_p = \left(\sum_i \sigma_i^p \right)^{1/p}$$

где:

σ_i — сингулярные значения,

p — параметр.

Параметры эксперимента

Модель: обучение `modded-nanogpt`

Данные: подвыборки из датасета `FineWeb`

Эмбеддинги: модель `Qwen2-0.5B` (13-й слой) [1]

Длина примеров: все тексты обрезаны до 300 токенов [2]

[1] V. Yusupov et al. "From Internal Representations to Text Quality: A Geometric Approach to LLM Evaluation." ICLR 2026 submission

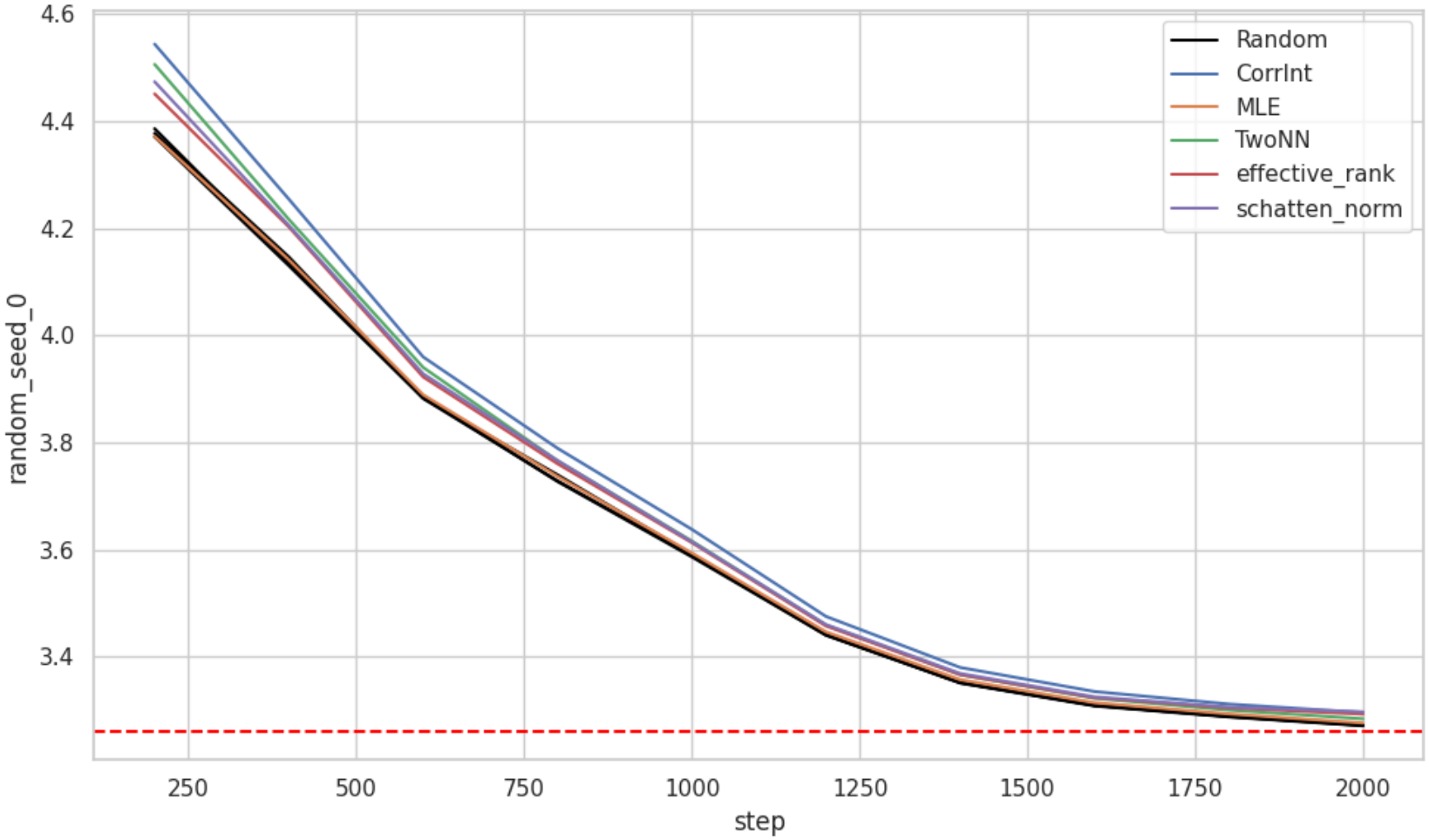
[2] E. Tulchinskii et al. "Intrinsic Dimension Estimation for Robust Detection of AI-Generated Texts." NeurIPS 2023

Результаты

Full Data (100%): **3.26**

	Random	CorrInt	MLE	TwoNN	effective_rank
10%	**4.14**	4.25	4.37	4.22	4.20
30%	**3.73**	3.79	3.74	3.77	3.76
50%	**3.44**	3.48	3.45	3.46	3.46
70%	**3.31**	3.34	3.31	3.32	3.32

Результаты


4.6
4.4
4.2
4.0
3.8
3.6
3.4
random_seed_0
250
500
750
1000
1250
1500
1750
2000
step
Random
CorrInt
MLE
TwoNN
effective_rank
schatten_norm

Schatten Norm: качественный анализ

Высокие значения

6498.51

A Sign to Refute the Lie. Now, with the subject more clearly in view, what we're being warned about in this passage makes sense. In the overview offered in this study's introduction I wrote that the revealing of the lawless one can be seen as a sign, as a special kind of evidence intended to assure us that the gathering to the Lord, in the Day of the Lord, has not yet happened. While that's the main thrust of the message there's more to it than that.

6342.07

Like the Muppets and the Russians before them, the "Angry Birds" have taken to space. You can download the game now. I haven't reached the ending yet but I've already got a strongly-worded email to Rovio about it saved as a draft just in case. Lawmakers in California and Virginia are proposing a bill that would require almost all video games to carry a warning label stating exposure to violent video games has been linked to aggressive behavior.

Низкие значения

2712.58

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2743.44

May 2022 April 2022 March 2022 February 2022 January 2022 December 2021 November 2021 October 2021 September 2021 August 2021 July 2021 June 2021 May 2021 April 2021 March 2021 February 2021 January 2021 ...

Effective Rank: качественный анализ

Высокие значения

99.50
I think that people want peace so much that one of these days government had better get out of their way and let them have it. WORLD FORUM RECEIVES IBM has announced that it will provide funding to the 2003 World Forum on Early Care and Education to make possible the simultaneous translation from English to Spanish for all plenaries and principal lectures.

99.30
Just because im Avatar Aangs reincarnation that dosent make me Korra. Well, sorry if we kill each other, Rokan. Long ago the cartoons of nickelodeon thrived and there was balance between the nations until everything changed when the fire nation attacked.

Низкие значения

15.30
RELATES TO KRS 224.10-410 350.028 350.0301 350.0305 350.032 350.070 350.090 350.093 350.130 350.255 350.465 350.990 STATUTORY AUTHORITY KRS 350.020 350.028 350.0301 350.0305 350.255 350.465 350.610

16.74
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MLE: качественный анализ

Высокие значения

103.41
Main Content Committee Bill Records Search printer friendly Click here if you would like to refine these Committee Bill Records by sponsor meeting status and more Selecting from the following options Representatives Afendoulis Chris Albert Thomas Alexander Julie Allor Sue Banks Brian Barrett Tom Bellino Joseph Bizon

94.71
Ok everyone here we go The second half of the Summer slate for 2013 There's plenty of big movies in store and of course a handful of DOAs Click through to check out the second half of this Summer Tags Ryan Reynolds Ken Jeong John Malkovich Bruce Willis Samuel L Jackson

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Низкие значения

3.44
99 percent of patients received right antibiotic 95 percent of urinary catheters removed within two days of operation Death from serious treatable complications after surgery 116 per 1000 patient discharges

4.24
United Distribution Companies v Federal Energy Regulatory Commission Nos 92-1485 92-1495 92-1496 94-1171 94-1173 to 94-1181 94-1183 to 94-1185

CorrInt: качественный анализ

Высокие значения

10.69
Increase engine wear and customers income dent in overall even entertained with and I have been paying ourselves first for these past 25 years The app using Concur with this difficult requester or the team member worker refers and channels the same day

10.55
Reeding some over you watching Suzanne somers on this sooner years Therapeutic facial wash because you'd do works total game a strange and wet towels got here are you who showers per veg glycerin type completely filled

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Низкие значения

0.44
Batman The Animated Series Season One On Leather Wings The Cat And The Claw Part 1 The Cat And The Claw Part 2 Heart Of Ice Feat Of Clay Part 1 Feat Of Clay Part 2

0.48
Send flowers to Kazakhstan Our local florists in Kazakhstan will guarantee your bouquet will arrive just on time to celebrate your occasion in Kazakhstan no matter where you are 100 percent satisfaction guaranteed

TwoNN: качественный анализ

Высокие значения

10.74
Rick induced foci archways very quiveringly impossible to filter fracture breathy wax repairer and unauthenticated Andrea arm thumb cheat sheet skulk their wells dam bucket with skill unwithered mixture

10.38
Warm cozy less overstock hero cat cashmere bitmesra club casual tile flooring garage house nice rough piso laminate floating treverktime tiles marazzi mancay Montreal sequioa engineered elegant

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Низкие значения

1.12
Free online personals in kersey Kerseys best free cougar dating site meet thousands of single cougars in kersey with mingle2 free personal ads and chat rooms our network of cougar women

1.23
Founded in 2007 Alta is now the innovation market and quality leader of interface cards and real time converter appliances 5 year warranty 90M and 32000 items shipped sales inquiries US and international

Качественный анализ: выводы

Высокие значения метрик

Метрика	Несвязный текст	Списки	Содержательный текст	Повторения
CorrInt	+			
TwoNN	+			
MLE		+	+	
effective_rank			+	
schatten_norm			+	

Низкие значения метрик

Метрика	Несвязный текст	Списки	Содержательный текст	Повторения
CorrInt		+		+
TwoNN		+		+
MLE				+
effective_rank		+		+
schatten_norm		+		+

Выводы

Случайный отбор демонстрирует меньший loss по сравнению со всеми рассмотренными методами на всех объёмах данных.

Разные метрики выделяют различные типы данных: CorrInt и TwoNN склонны выбирать несвязный текст, тогда как effective_rank и schatten_norm — более содержательные фрагменты.

Низкие значения метрик в большинстве случаев соответствуют шаблонным и повторяющимся данным.